A Tool for Furthering GNSS Security Research: The Oak Ridge Spoofing and Interference Test Battery (OAKBAT)

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BIOGRAPHIES

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ABSTRACT

A new global navigation satellite system (GNSS) dataset of digitized RF signals named the Oak Ridge Spoofing and Interference Test Battery (OAKBAT) has been created. OAKBAT contains digitized spoofing signals that serve as both a supporting "sibling," and an advancement to the widely used Texas Spoofing Test Battery (TEXBAT) dataset [1]. OAKBAT at its core was developed to 1) allow for 100% reproducibility of the data, 2) provide more detailed metadata and contextual information about each dataset such as the precise moment the spoofing and/or interference signals begins, positions used, constellations visible, etc., and 3) to provide datasets generated using the same key parameters as the TEXBAT ds1 through ds6 datasets. Through these "sibling" datasets it will now be possible to determine if the behavior of algorithms and designs developed utilizing these datasets are affected by possible intrinsic behavior and properties of the equipment used in the generation and digitization of either of the two datasets. The end goal of OAKBAT is to be a new resource for the community of researchers and developers working in the field of GNSS. This additional source of data provides a GNSS "playground" on which to experiment, explore, and evaluate new ideas.

INTRODUCTION

Motivation

Although it has been some time since the public release of the Texas Spoofing Test Battery (TEXBAT) [1], TEXBAT continues to be incredibly useful for those working in the areas of security and robustness of global navigation satellite systems (GNSS) receivers. The primary downside to the TEXBAT dataset is its small size: only eight sets. In order to more thoroughly evaluate potential algorithms and designs, additional data is needed, making it necessary to either purchase a commercial GNSS simulator system or build a system capable of creating spoofing, jamming, and various interference signals. The commercial systems carry a non-trivial financial cost. Likewise, the cost in time, money, and the potential for mistakes when building a sophisticated spoofing device from the ground up, which is capable of reproducing at a minimum the same scenarios as contained in TEXBAT, is also high. These costs are felt most severely by newcomers to the field and those going out on their own academic or commercial endeavors. These challenges, and the ways in which they have inhibited our own research, motivated us to create and release to the public a new dataset of GNSS spoofing signals.

This new dataset is named the Oak Ridge Spoofing and Interference Test Battery (OAKBAT). This collection of digitized RF signals has been created as both a supporting "sibling" and an advancement to the widely used TEXBAT dataset. The scenarios that make up this initial release of OAKBAT were developed with several key characteristics and goals in mind for the benefit of the GNSS research community at large. First, the datasets need to be 100% reproducible by an independent party. Second, more detailed metadata and contextual information such as the precise moment the spoofing and/or interference signals begin, the location used for static position files, the date and time used, the route/path of dynamic files, the GNSS constellations present, and the satellite vehicles in view must be provided for each dataset. Third, to produce datasets which mimic the key parameters of the TEXBAT ds1 through ds6 and the associated spoofing-free baseline sets as closely as possible. For OAKBAT these six "twin" sets, along with the two baseline clean scenarios they are built from, are recordings of the GPS L1 C/A signal. However, OAKBAT includes an addition six sets and two clean baseline sets that are recordings of Galileo E1 signal. These Galileo OAKBAT recordings are identical to the OAKBAT GPS L1 C/A recordings in every parameter other than the GNSS constellation.

In particular, the goal of independent reproducibility permeates the OAKBAT dataset. It stems from our own experience working with "black box" data. Without knowing exactly where and how a set of data was acquired or created often leaves one questioning any unexpected outcomes resulting from its analysis and/or usage. By providing users with the exact equipment, configuration, scenario files, and parameters used to generate and digitize the data they may further investigate and determine the root of any unexpected results and even build upon this information to create their own datasets.

The support of reproducibility necessitates providing more information and detailed metadata on each dataset to be released. OAKBAT benefits in its ability to supply detailed metadata due to the fact that the data is generated using a commercially available GNSS simulator. OAKBAT has been produced using an Orolia Vulnerability Test System (VTS) [2] and therefore the "metadata" for each recording includes the VTS simulator configuration files used in the generation of the digitized signals. Descriptions of the exact equipment used for the generation and digitization follows.

EQUIPMENT CONFIGURATION

Multi-GNSS Simulator

The RF source for the datasets is a commercially available product, specifically designed and assembled to allow the production of signals such as interference, multipath, and spoofing of both single and multiple GNSS constellations, just to name a few of the possible signal generation capabilities. The exact system used is the Orolia VTS. It consists of two Orolia GSG-6 series multi-frequency, multi-constellation capable GNSS simulators. The simulators are connected to an Orolia SecureSync time and frequency source which is also a Network Time Protocol (NTP) server. The SecureSync is configured to serve NTP, provide a 10 MHz reference signal, and a 1 pulse-per-second (PPS) time reference all synchronized and referenced to the Rubidium oscillator contained in the SecureSync. The installed Rubidium source has an accuracy of 0.001 ppb.

The NTP server is utilized by the GNSS simulators to both set the date and time simulated and to synchronize the generation of the RF signals of the two units. The Rubidium frequency reference is operated in a free-running mode to allow the user to manually set the date and time at will. The entire VTS is frequently reconfigured so the long-term consequences of using an undisciplined Rubidium reference are not applicable. Additionally, the fact that the OAKBAT datasets are only eight minutes in duration and that the VTS is powered off at the end of each workday should remove any meaningful concern over the use of an undisciplined Rubidium oscillator.

The two GNSS simulators' RF outputs are combined using a GPS Source C21 combiner, which is part of the VTS. The resulting combined signal is the output of the VTS. The signal strength of each simulator has been independently calibrated to account for the cabling losses and loses in the combiner, so that power set in simulation scenarios or via the instrument interface(s) is the power seen at the RF output of the VTS (post-combiner). The simulator as a whole is controlled using the Orolia GSG StudioView software which is operated from a computer connected to the VTS's built-in ethernet switch. This software allows reusable scenarios to be created that define the starting location, routes, GNSS constellations and signals, ionosphere model, propagation environment, and much more. The VTS allows the use of a GSG StudioView option known as the Spoofing Tool which enables the user to select the scenarios used for the sky and the spoofer, power offsets, spoofing type and amount, etc. and is responsible for loading, synchronizing, and initiating the two GSG-6 simulators at precisely the same moment.

Regardless of the commercial entity supplying the simulator, it is the complexities and intricacies, as well as the ability to consistently repeat the process, that make the use of a commercial GNSS simulator with spoofing and interference capabilities a beneficial asset. At the same time, it is the complexities, cost in time, effort, as well as financial cost of these task and the hardware capable of performing them, which makes OAKBAT and TEXBAT a beneficial stepping stone in the development and evaluation of a new idea without incurring significant financial and material resources typically unavailable to ideas and research in its infancy.

Recording Equipment

The acquisition equipment selected to digitize the signals from the GNSS simulator was chosen based on cost and availability. The system used for both recording and playback of the recorded files was a software defined radio (SDR) by Ettus Research, the X310 with a WBX-40 front-end card. The X310 is a mainboard/carrier for a variety of RF front ends, commonly referred to as daughter cards. The X310 contains a quadrature pair of 14-bit analog-to-digital convertors (ADC) and a quadrature pair of 16-bit digital-to-analog convertors (DAC) [3]. The WBX-40 has a 40 MHz bandwidth for both receive and transmit. It is capable of being tuned to any frequency from 50 MHz up to and including 2.2 GHz [4]. In order to digitize the RF signal with more resolution a pair of low noise amplifiers (LNAs) were used to amplify the signal so as to use more of the ADCs bit-depth. The two LNAs and their key properties are listed in Table 1. The first of the two LNAs was selected specifically to set the noise figure of the full receive chain. The second LNA was selected for its gain.

Table 1. Low noise amplifiers and their key parameters used during the RF digitization recording process.

Manufacturer	Model	Gain @ Frequency	Noise Figure @ Frequency
Mini-Circuits	ZX60-P33ULN+	14.25 dB @ 1550 MHz	0.42 dB @ 1550 MHz [5]
Mini-Circuits	ZX60-2534MA-S+	43.42 dB @ 1500 MHz (5V)	2.24 dB @ 1500 MHz (5V) [6]

In addition to the 57 dB of external gain used, the WBX-40 front-end has up to 31.5 dB of internal analog receiver gain available of which all 31.5 dB was utilized while recording the OAKBAT data. The performance data for the WBX daughterboard reports a noise figure of approximately 5 dB at 1500 MHz. Calculating the cascaded noise figure and gain for the two LNAs plus the WBX results in a noise figure of 0.52 dB and a total gain of 89 dB. GNU Radio was used as the controlling software for the SDR.

The X310 SDR supports the use of both an external frequency reference input and an external PPS clocking input. The SecureSync timing source integrated into the VTS provides a sufficient number of output ports so that both the 1-PPS and the 10 MHz reference were supplied to the two GSG-6 simulators and the SDR without needing to share a port, i.e. no delay inducing signal splitters were needed. This allows both the GNSS simulators and the SDR to have a shared common frequency and timing reference which provides excellent sample clock synchronization between the signal generators and the signal digitizer.

Recording Parameters and Data Format

The RF signal produced by the simulation system was sampled using a 5 MHz sampling rate and since the SDR uses a quadrature sampler the bandwidth of the recorded signal is 5 MHz as well. The digitized RF data is saved in complex interleaved format using an individual int16 datatype for the in-phase value followed by a second int16 holding the quadrature value. This format is known in GNU Radio terminology as "IShort", i.e., interleaved-short form. As a side note, this data format is the same datatype and format as used by the TEXBAT data files. The only change needed to adapt existing work to use the OAKBAT data files instead of the TEXBAT data files is to adjust for the difference in the sampling rates.

GNSS Simulator and Recording Systems Interconnect

The complete layout of the GNSS simulator, including its sub-components, along with the SDR and its peripherals as assembled and used during the recording of the OAKBAT scenarios is shown in Figure 1. All interconnects are shown in the diagram with the exception of the power connections.

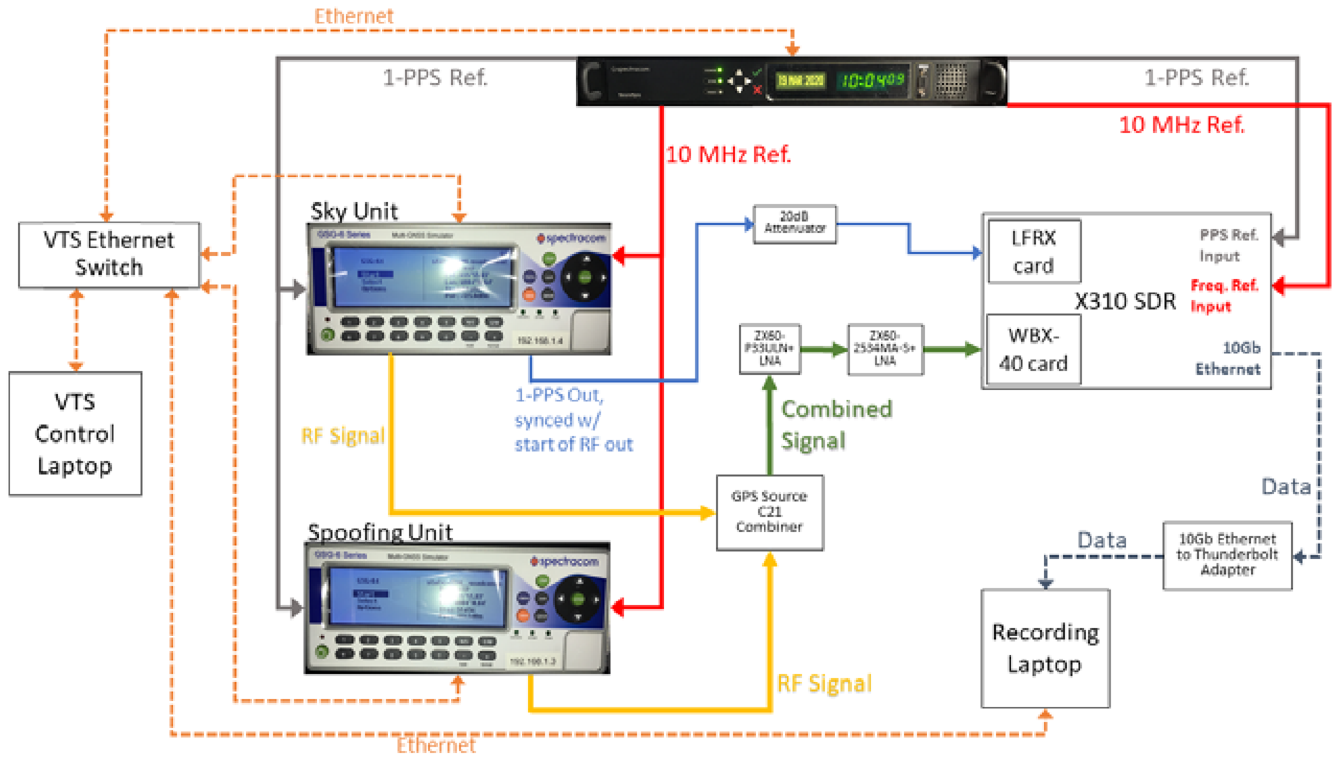

Figure 1. The complete simulation system and recording system configuration diagram. The power cables alone have been excluded from the diagram.

DATASETS

OAKBAT consist of 16 unique datasets. Eight sets contain only the GPS L1 C/A signal, two being the spoof-free, clean baseline sets and the other six containing various degrees and types of spoofing. Another, eight sets contain only the Galileo E1 signal, two sets containing the spoofing-free, clean baseline sets while the remaining six containing various degrees and types of spoofing.

Common Scenario Attributes

There are several key parameters common across all scenarios as well as two parameters that are common to the main categories of scenarios, i.e., static, dynamic, GPS, and Galileo. The first and most important common parameter is the starting date and time, March 19, 2020 9:59:42 UTC (March 19, 2020, 10:00 GPS Time). All the scenarios and thus recordings begin at this precise date and time. The second is the environmental and atmospheric models used during the simulations of all scenarios. The propagation environment used is the “open-sky” option of the Orolia GSG-6 simulators. This environment uses the models specified in the ITU-R Recommendation M.1225, Annex 2, Part 2, Satellite component [7]. For detailed information on the ITU propagation models as used in the GSG-6 simulators, which are beyond the scope of this document, refer to Section 3.5.9.4 Environment Models in [8]. Additionally, the simulator’s default ionosphere model was enabled, which per Section 3.5.9.5 of [8] is “*a reverse model of the model described in IS-GPS-200D, Section 20.3.3.5.2.5, called Klobuchar.*” The same section of the GSG user manual also describes the details of the Saastamoinen tropospheric model. A “zero model” isotropic perfect sphere antenna gain pattern model with 0 dBic gain in all directions was used so as to have no impact on the generated signals. The final common attribute used was the target signal strength of the GPS satellites and the Galileo satellites at a receiver. The power level for the GPS satellites was set at -125 dBm while for the Galileo satellites a power level of -123.5 dBm was used. In the case of the open-sky propagation environment, no additional attenuation is applied to these levels. Finally, the simulators utilized the actual broadcast ephemeris data files for March 19, 2020 as acquired from the NASA Crustal Dynamic Data Information System (CDDIS) [9] for the GPS transmissions and the CNES Navigation and Time Monitoring Facility (NTMF) [10] for the Galileo transmissions.

The static receiver scenarios use the starting position of, 35° 55’ 49.96” N, 84° 18’ 38.35” W, with a starting altitude of 248.6-meters (WGS84) which is the center of the “quad” at Oak Ridge National Laboratory as shown in Figure 2. The initial GPS satellite constellation for this position at the starting time and date is shown in Figure 3. Likewise, the Galileo constellation as seen from this location at the same time and date is shown in Figure 4.

Figure 2. Static scenarios position. The center of ORNL Quad (35° 55’ 49.96” N, 84° 18’ 38.35” W). Imagery and map data from Google.

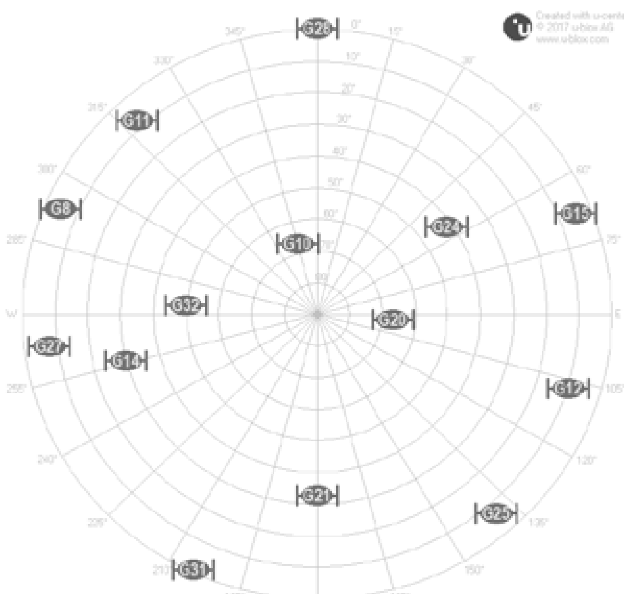

Figure 3. GPS constellation as seen from static position (ORNL Quad) at scenario starting time.

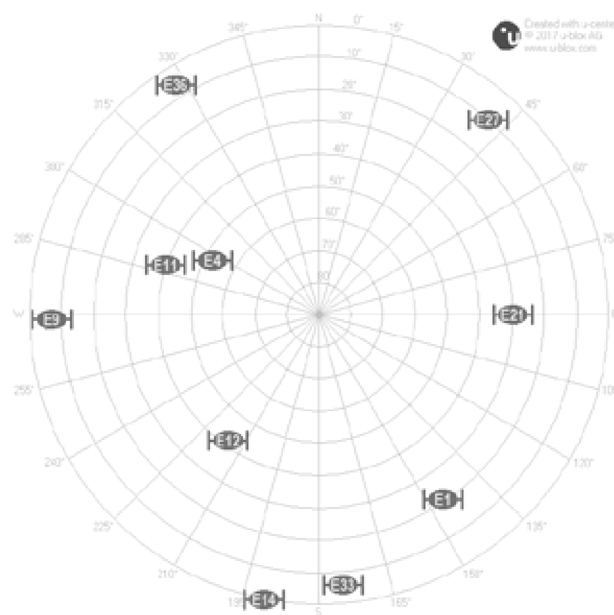

Figure 4. Galileo constellation as seen from static position (ORNL Quad) at scenario starting time.

For the dynamic moving, receiver scenarios, the route traveled is shown in Figure 5. This route travels from the American Museum of Science and Energy (ASME) to the Children's Museum of Oak Ridge. The velocities are simulated based on the actual speed limits. A smoothing and interpolation step was applied to the route to prevent unrealistic angular and vertical velocity changes and acceleration as the route is not from an actual GNSS receiver driven between the two museums. The simulated route was used to achieve a precisely 8-minute long recording which starts with one minute of stationary "lock" time in the AMSE's parking lot before a 7-minute drive to the Children's Museum parking lot. The 7-minute drive includes realistic dwell times at stop signs and traffic lights requiring a stop, such as a left turn. Traffic laws in the state of Tennessee allow a right turn at a stop light after coming to a full stop and there is one such stop in this route at the intersection of West Outer Drive and North Illinois Avenue as seen on the left-hand edge of Figure 5.

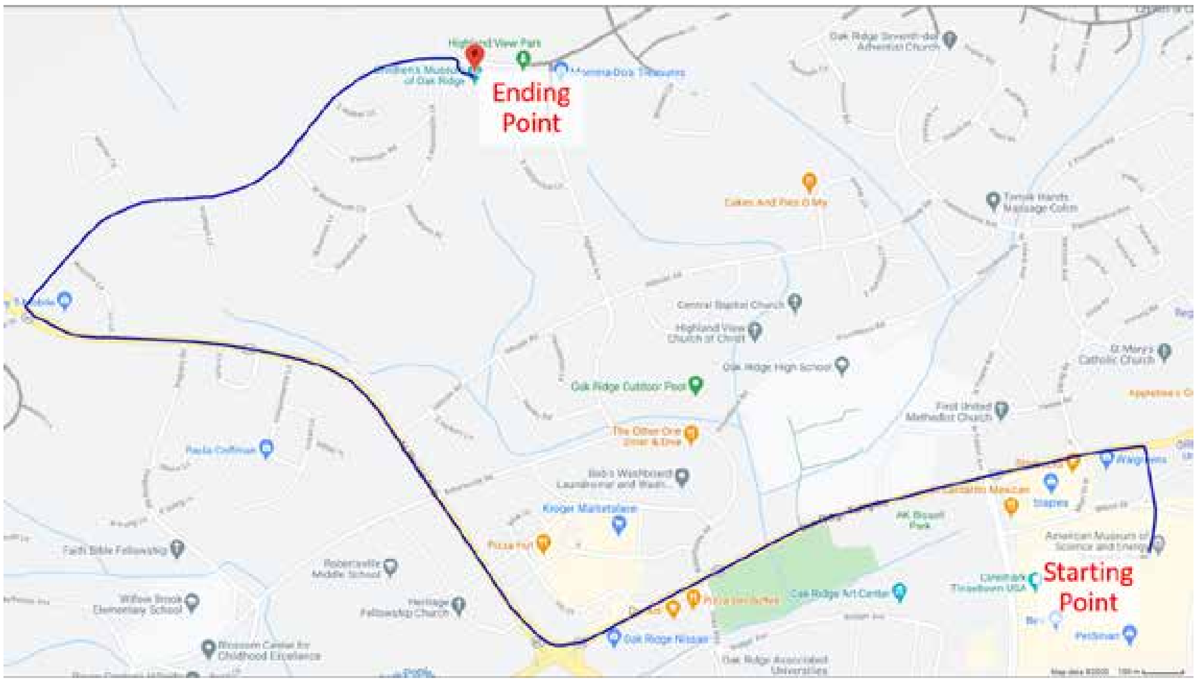

Figure 5. Route used for the dynamic scenarios. The starting location is the AMSE's parking lot (N35 00' 45.47", W84 15' 11.12", 271.4m (WGS84). The ending location is the parking lot of the Children's Museum (N36 01' 23.97", W84 16' 18.06", 337.8m (WGS84).

The GPS constellation as seen from the starting point of the dynamic scenarios at the OAKBAT starting time, March 19, 2020 9:59:42 UTC, is shown in Figure 6, while the Galileo constellation as seen from the same starting location, date, and time is shown in Figure 7.

Recorded Scenario Validation

To verify the contents of each RF recording a simplistic "quality" check of the digitized RF recordings was performed to ensure that the digitized recordings could be played back and a free-standing GNSS receiver could still achieve a position fix. The recordings were replayed using the X310 SDR and the WBX-40 daughterboard. The digital-to-analog convertor of the X310 has a 16-bit resolution depth. The 10 MHz Rubidium reference oscillator output of the VTS's SecureSync was supplied to the SDR as a reference oscillator. This reference is available from the VTS regardless of whether the system is in use, which it was not during these quality checks. No transmit gain was used during the play-back.

A u-blox EVK-M8T GNSS receiver evaluation kit containing a NEO-M8T receiver was used. The output of the SDR was cabled to a Mini-Circuits DC-block connected directly to the EVK-M8T's antenna port. The u-blox u-center software [11] was used to monitor and log a number of key parameters as determined by the receiver in response to each replayed RF recording. This was not an evaluation of the receiver's performance against spoofing or of the spoofing's effectiveness. It was an evaluation of the digitized signals and their ability to reproduce the recorded signal with enough fidelity to allow a common commercial GNSS receiver to achieve and hold a fix.

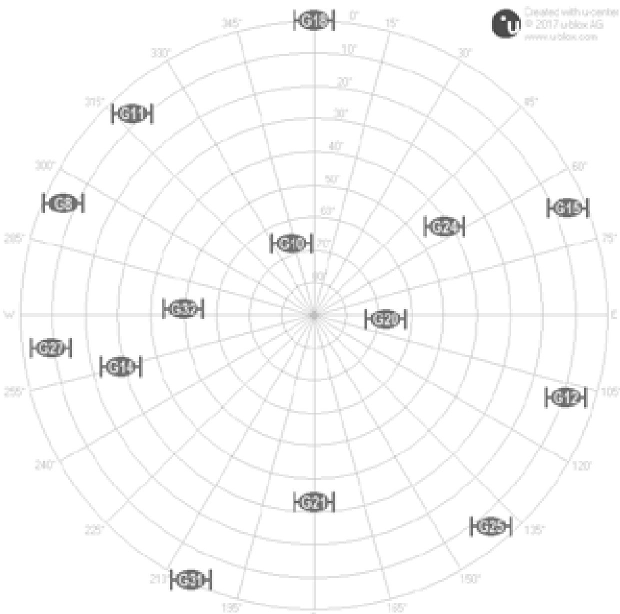

Figure 6. GPS constellation as seen from the starting position (AMSE's parking lot) of the dynamic scenario's route.

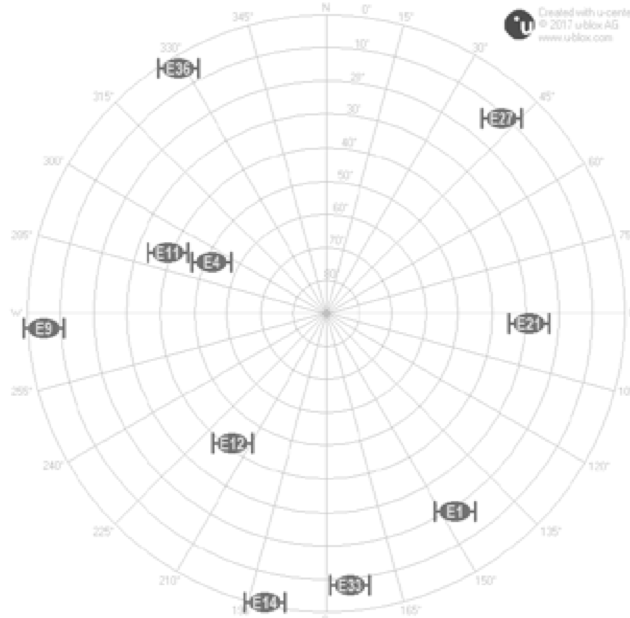

Figure 7. Galileo constellation as seen from starting position (AMSE's parking lot) of the dynamic scenario's route.

Scenario 1a: Static RF Switch – os1a

This scenario is more of an extended relative to the TEXBAT ds1 scenario than a sibling, hence the name "1a". In this scenario, spoofing is present in a more substantial way than only switching between the sky signal and an identical signal originating from the spoofer instead of over-the-air. Exactly mimicking the "attack" as contained in the TEXBAT ds1 set was not informative or noticeable due to the perfect synchronization between the two GNSS simulators existing in the Orolia VTS. Every aspect of the signal generated by the sky unit and the spoof unit is identical and synchronized, so without reducing the power of the signal or creating a time or position push, there is no detectable difference between the signal produced by the sky simulator versus the spoofing simulator. The scenario does mimic the "spirit" of TEXBAT ds1 scenario and is described below.

This static RF switch scenario emulates an event in which the attacker has physical access to the receiver's antenna signal and switches the receiver's incoming antenna signal from the true signal to the spoofer's signal by some near instantaneous means. The power levels, satellites present, etc. are identical between the sky signal and the spoofed signal. The difference between the sky and spoofing signal is that the spoofing signal is for a position 600-meters away from the sky's position in the horizontal plane as shown in Figure 8.

The difference between Earth centered, Earth fixed (ECEF) position as calculated by the u-blox EVK-M8T GNSS receiver used during the "quality check" of the os1a recording, as described in the Recorded Scenario Validation section, and the "perfect" ECEF coordinates for the static scenario's starting position are plotted for X, Y, and Z ECEF axes in Figure 9.

The "perfect" ECEF coordinates are the ECEF coordinates as converted from the starting Latitude, Longitude, and height above the WGS84 ellipsoid used in the scenario files for the VTS unit. Since we have these exact locations as used in the simulator, it is a simple matter to convert them to the "perfect" ECEF coordinates. Similarly, since the satellite most directly overhead is easily known thanks to the use of the GNSS simulators and the satellite preview capability in the GSG StudioView software, we show the carrier-to-noise ratio (CN_0) of the "best" satellite in Figure 10. In the case of the GPS scenarios the "best" satellite is GPS SV G10, see Figure 3, while for the Galileo scenarios the "best" satellite is Galileo SV E4, see Figure 4.

Figure 8. Static positions used in “os1a” RF switch scenario.

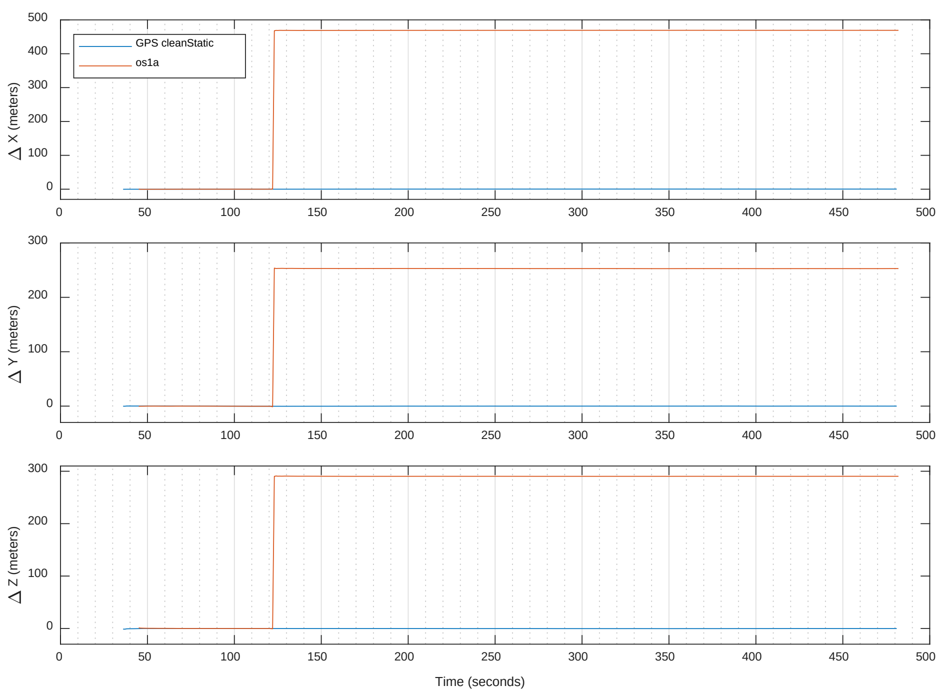

Figure 9. Difference between the “perfect” ECEF coordinates for the static position scenario and the ECEF coordinates calculated by the u-blox EVK-M8T receiver for Scenario 1a.

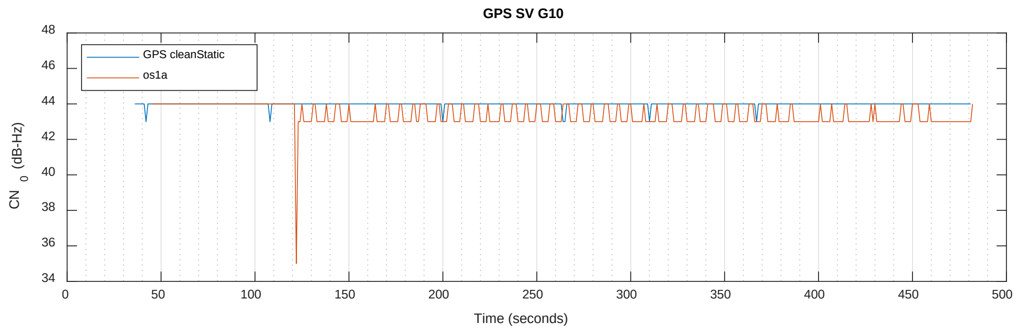

Figure 10. Scenario 1a's CN_0 as calculated by the u-blox EVK-M8T for the "best" positioned GPS satellite G10. "Best" refers to the satellite most directly overhead, see Figure 3.

Scenario 2: Static Time Push, 10 dB Spoofing Power Advantage – os2
Scenario 2 simulates the GPS constellation as seen by a statically located receiver positioned at the center of the ORNL quad, Figure 2. A two-microsecond time push begins at the 2-minute mark from the start of the recording. This scenario is a "twin" to the TEXBAT scenario 2 (ds2) with respect to the key parameters of the scenario. In this scenario, the spoofing signal has an overwhelming power advantage at 10 dB higher power than the sky signals. The difference in the ECEF coordinates derived by the u-blox EVK-M8T with respect to the "perfect" ECEF coordinates calculated using the exact location defined in the StudioView scenario file are shown in Figure 11. Figure 12 shows the CN_0 as calculated by the EVK-M8T for the GPS G10 satellite vehicle, which is the satellite closest to being positioned directly overhead, see Figure 3.

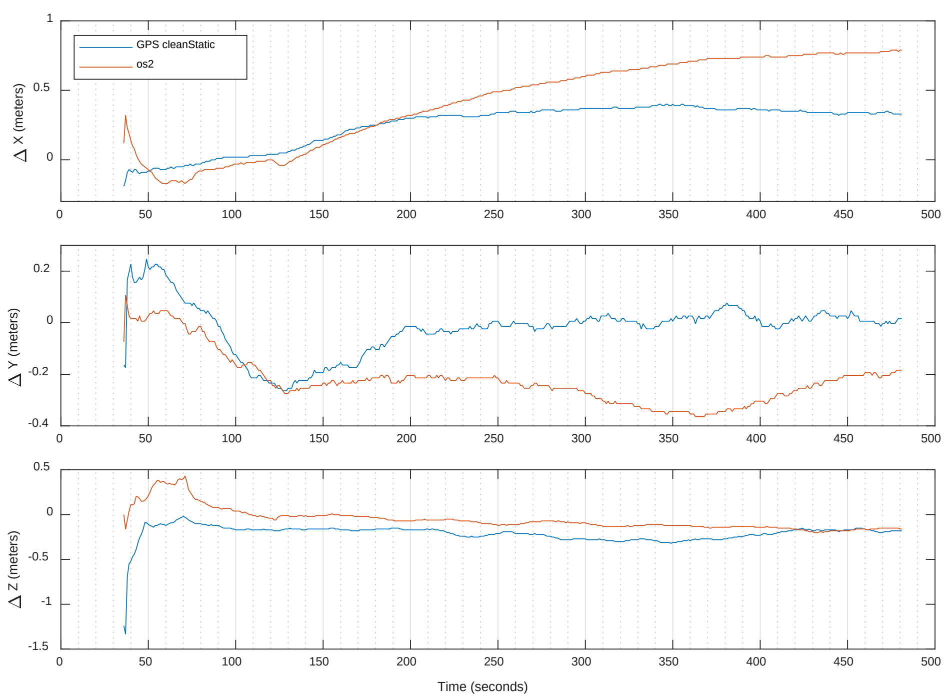

Figure 11. Difference between the "perfect" ECEF coordinates for the static position scenario and the ECEF coordinates calculated by the u-blox EVK-M8T receiver for Scenario 2.

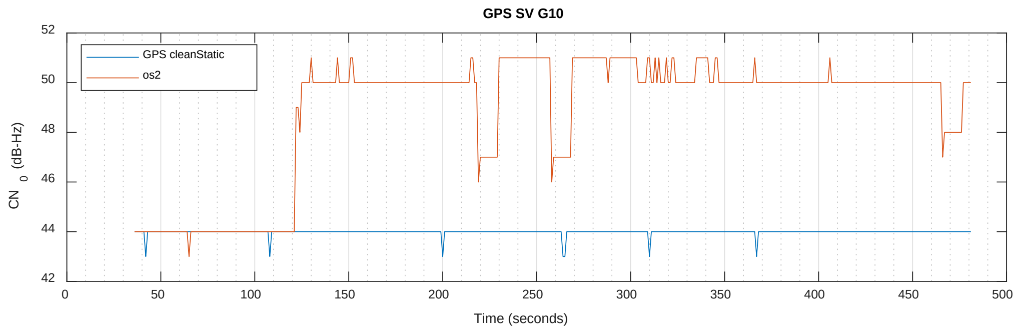

Figure 12. Scenario 2's CN_0 as calculated by the u-blox EVK-M8T for the "best" positioned GPS satellite G10. Where "best" refers to the satellite most directly overhead, see Figure 3.

Scenario 3: Static Time Push, 1.3 dB Spoofing Power Advantage – os3

Scenario 3 is identical to Scenario 2 apart from the level of the power advantage the spoofing signal has over the sky signal, at only 1.3 dB of power advantage compared to Scenario 2's 10 dB advantage. This scenario is a "twin" to the TEXBAT scenario 3 (ds3) with respect to the key parameters of the scenario. As before, the difference in the ECEF coordinates with respect to the "perfect" ECEF coordinates are shown in Figure 13. The CN_0 as calculated by the EVK-M8T for the GPS G10 satellite vehicle, the satellite closest to being positioned directly overhead as seen Figure 3, is displayed in Figure 14.

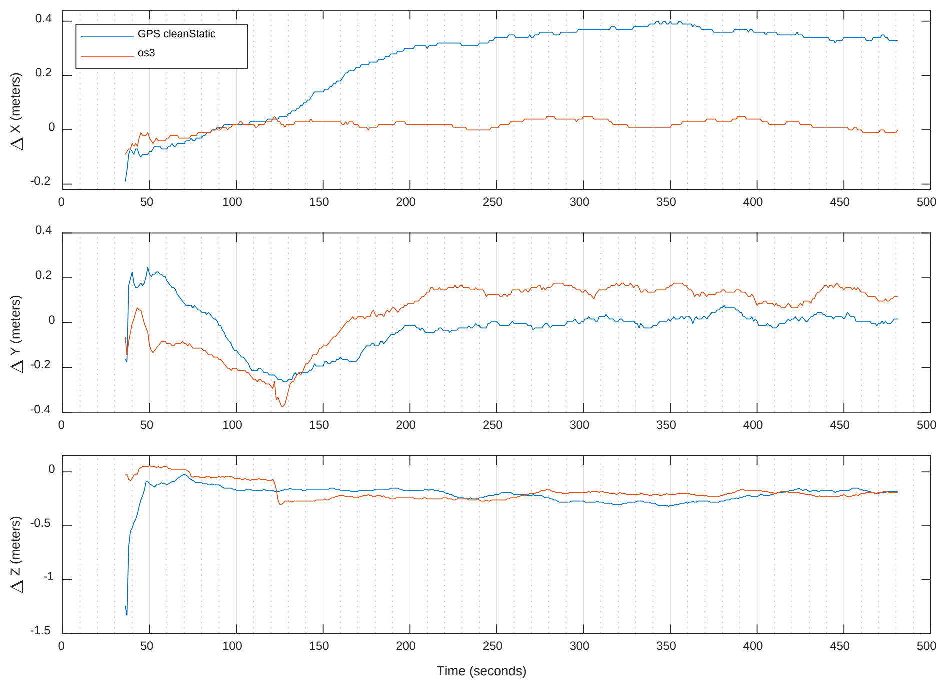

Figure 13. Difference between the "perfect" ECEF coordinates for the static position scenario and the ECEF coordinates calculated by the u-blox EVK-M8T receiver for Scenario 3.

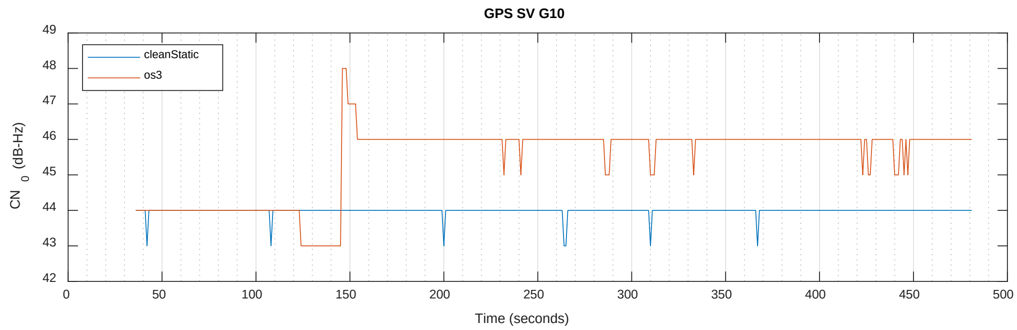

Figure 14. Scenario 3's CN_0 as calculated by the u-blox EVK-M8T for the "best" positioned GPS satellite G10. Where "best" refers to the satellite most directly overhead, see Figure 3.

Scenario 4: Static Position Push, 0.4 dB Spoofing Power Advantage – os4
Scenario 4 simulates a static receiver located at the center of the ORNL quad, as shown in Figure 2, that is exposed to a position push instead of a time push. The power advantage of the spoofer is quite weak in this case at only 0.4 dB greater than the sky. The position push is a 600-meters push in the ECEF Z-direction with respect to a receiver's perceived location.. This scenario is a "twin" to the TEXBAT scenario 4 (ds4) with respect to the key parameters of the scenario. Keep in mind that one of the purposes of OAKBAT was to create a second version of the TEXBAT scenarios to benefit the GNSS community as a whole through the additional data representing the same scenarios yet produced by two different sets of hardware. As such, the effectiveness of a particular spoofing attack or the ability of a receiver to resist an attack is not the focus of this work.

Therefore, with these things in mind, Figure 15 shows the change between the ECEF coordinates calculated by the u-blox EVK-M8T and the "perfect" ECEF coordinates for the static location exhibit small drifts and the change in the z-axis is quite small. It is clear, that the drifts in the ECEF coordinates do coincide with the onset of the spoofing signal at the 120-seconds mark. In the same vein, Figure 16 shows the CN_0 as calculated by the EVK-M8T for the GPS satellite located in the "best" position, GPS G10, which exhibits a small reduction in power coinciding with the onset of spoofing. The difference between the average of the calculated CN_0 for Scenario 4 once spoofing begins and the CN_0 of GPS G10 calculated from clean static baseline file is 1.5 dB. Upon closer examination, it is noticeable that the EVK-M8T only reports CN_0 values using a granularity of 1 dB. It may seem counter intuitive to see the reduction in CN_0, but it is "flutter" specific to the GNSS receiver used and the resolution of its reported CN_0 values. The reported power output of each GSG simulator was re-checked and found to be as expected, -125 dBm for the sky signals and -125.4 dBm for the spoofing signals (once spoofing had initiated).

Scenario 5: Dynamic Time Push, 9.9 dB Spoofing Power Advantage – os5
Scenario 5 simulates a dynamic receiver. An initial minute of stationary dwell time starts the route followed by a 7-minute drive as shown in Figure 5 and described previously. The spoofing attack is initiated exactly 2-minutes into the recording and is a 2-microsecond time push with a significant power advantage of 9.9 dB over the sky signal strength.

The ECEF coordinates for the dynamic scenarios were determined using a u-blox EVK-M8T GNSS receiver at a 1 Hz refresh rate as previously discussed in Recorded Scenario Validation section. To normalize the sets of ECEF coordinates the time stamp logged along with the coordinates for each recording was used to time align a scenario's ECEF coordinates with ECEF coordinates the EVK-M8T calculated for the clean-dynamic baseline. Then the clean-dynamic baseline's ECEF coordinates were subtracted from the scenario's coordinates. This step was also applied to the clean-dynamic coordinates, thus establishing the normalized baseline ECEF coordinate values in each ECEF axis as zero.

Figure 17 shows the normalized difference between the ECEF coordinates of the spoofing free clean dynamic recording and the normalized ECEF coordinates for Scenario 5. As confirmation that the spoofing is occurring the CN_0 of the "best" GPS satellite is plotted in Figure 18. The "best" satellite is defined as the satellite most directly overhead as shown in Figure 6. For OAKBAT GPS scenarios, the GPS constellation at the starting time for the dynamic route shows that GPS SV G10 is closet to directly overhead. The onset of the spoofing, especially given the significant power advantage is clearly visible at the 2-minute mark.

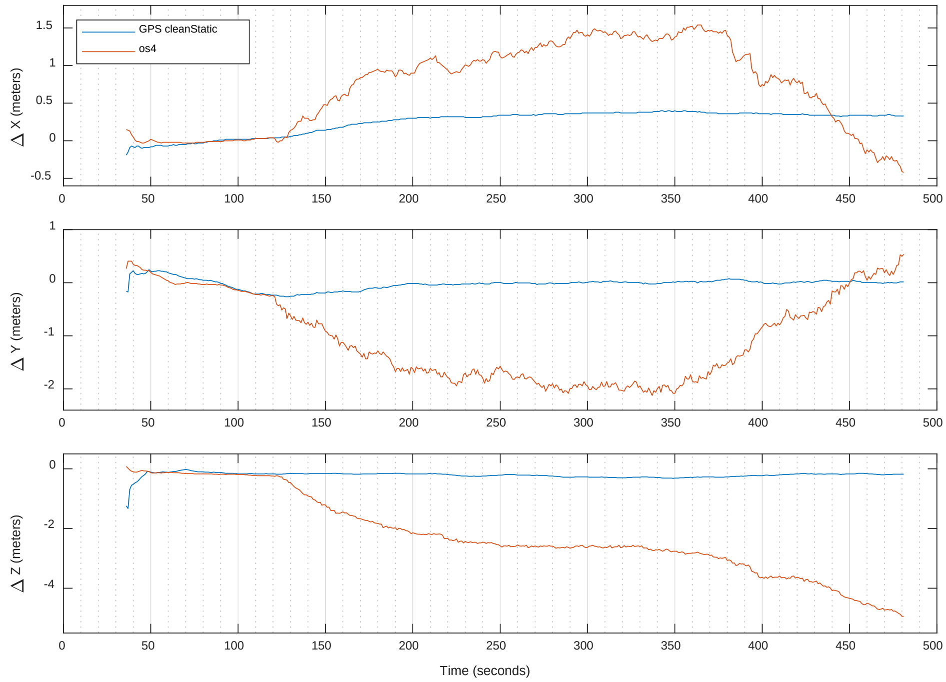

Figure 15. Difference between the "perfect" ECEF coordinates for the static position scenario and the ECEF coordinates calculated by the u-blox EVK-M8T receiver for Scenario 4.

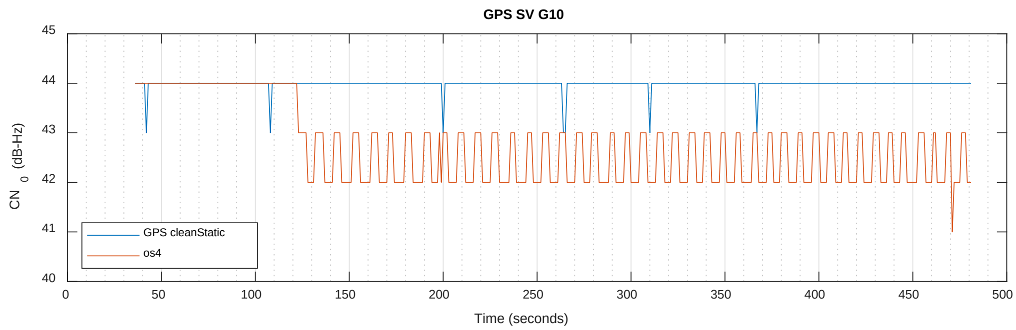

Figure 16. Scenario 4’s CN0 as calculated by the u-blox EVK-M8T for the “best” positioned GPS satellite G10. Where “best” refers to the satellite most directly overhead, see Figure 3.

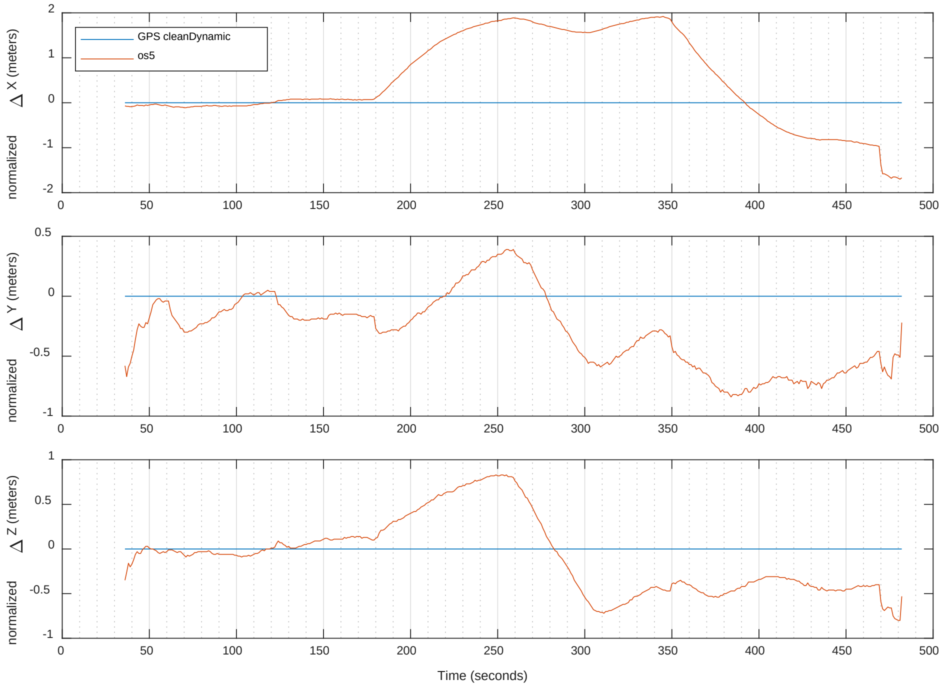

Figure 17. Difference between the normalized clean dynamic scenario's ECEF coordinates and the normalized ECEF coordinates for Scenario 5. Both sets of coordinates were derived using the u-blox EVK-M8T.

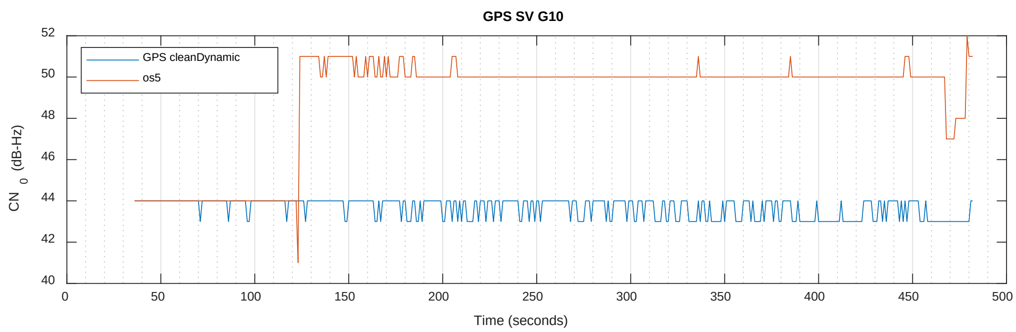

Figure 18. Scenario 5's CN_0 as calculated by the u-blox EVK-M8T for the "best" positioned GPS satellite G10, see Figure 6 as to why GPS SV G10 is the "best".

Scenario 6: Dynamic Position Push, 0.8 dB Spoofing Power Advantage – os6

Scenario 6 is the same as Scenario 5 with regards to the dynamic route traveled and the 1-minute stationary dwell at the beginning of the scenario. The only differences are that the spoofer's power advantage over the sky signal is only 0.8 dB and a 600-meter position push in the ECEF Z-direction with respect to a receiver's perceived location is spoofed instead of a time push. The time synchronized difference between the normalized clean dynamic baseline's ECEF coordinates, as calculated by the EVK-M8T GNSS receiver, and the normalized ECEF coordinates derived by the EVK-M8T for Scenario 6 are shown in Figure 19. As with the ECEF coordinate differences for Scenario 4, the differences between the spoofing-free clean dynamic scenario's calculated ECEF and Scenario 6's are not as one might expect. Still, the onset of spoofing is clearly visible at the 120-second mark and as previously stated, the effectiveness of the spoofing and susceptibility of the receiver used to determine the ECEF coordinates is not the focus of this document describing the OAKBAT dataset. Figure 20 shows a small reduction in CN_0 calculated by the EVK-M8T once spoofing has begun. This same small drop and flutter is similar to the CN_0 drop seen between the spoof-free static baseline recording and Scenario 4's CN_0 as seen in Figure 16. As with Scenario 4, the powers of both GSG simulator units were re-checked and found to be as expected via repeat runs. This small but meaningful "re-check ability" being possible thanks to the reliable, reproducibility that is part of the OAKBAT collection procedure and equipment.

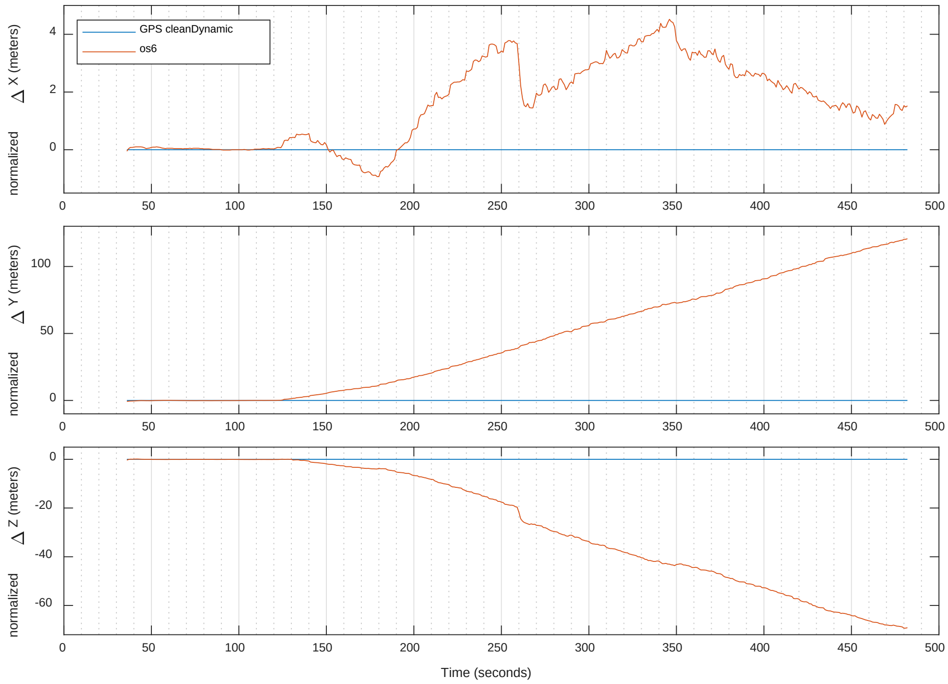

Figure 19. Difference between the normalized clean dynamic scenario's ECEF coordinates and the normalized ECEF coordinates for Scenario 6. Both sets of ECEF coordinates were derived using the u-blox EVK-M8T.

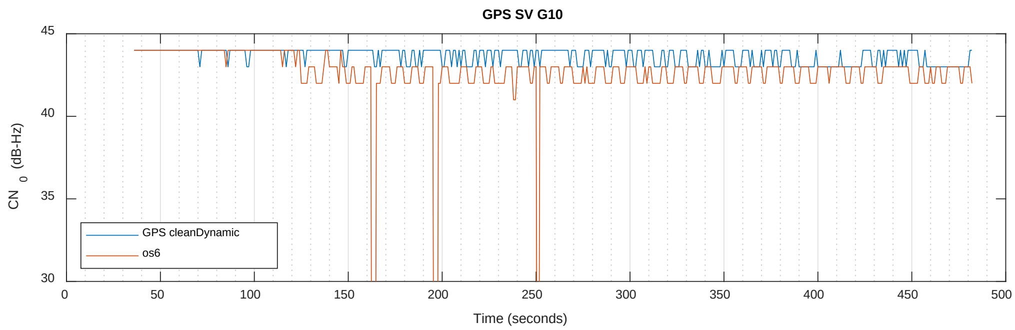

Figure 20. Scenario 6's CN_0 as calculated by the u-blox EVK-M8T for the "best" positioned GPS satellite G10, see Figure 6 as to why GPS SV G10 is the "best".

Scenarios 9a, 10, 11, 12, 13, and 14: Galileo Only Versions – os9a, os10, os11, os12, os13, os14

Scenarios 9a, 10, 11, 12, 13, and 14 are identical in every detail to the GPS only Scenarios 1a, 2, 3, 4, 5, and 6 except in one detail, instead of the GNSS signal being GPS L1 C/A it is Galileo E1. The rows of Table 2 explicitly show the correspondence between the GPS and Galileo scenarios. As is noticeable in Table 2, there is no Scenario 7 or 8 in the OAKBAT dataset. This is intentional as the complex phase and amplitude inversions along with the rate-based parameter changes of TEXBAT ds7 and ds8 were not conducive to reproduction and reliable repeatability using the available equipment. Therefore, to avoid any potential confusion and unintentional assumptions of equivalency between TEXBAT ds7 and ds8 there is no Scenario 7 (os7) or 8 (os8) in the OAKBAT collection.

The Galileo scenarios use the exact same locations, starting date and time, power parameters, etc. as the GPS only scenarios. To improve the readability of this document the description of each scenario, the ECEF difference plots, and the CN_0 plots for the six Galileo scenarios are available to interested readers on the metadata repository described in Open Data Access section below.

Table 2. Scenario equivalency between GPS only and Galileo only OAKBAT scenarios.

Scenario Equivalencies		
GPS Only	**Galileo Only**	**TEXBAT**
Scenario 1a (os1a)	Scenario 9a (os9a)	Scenario 1 (ds1)
Scenario 2 (os2)	Scenario 10 (os10)	Scenario 2 (ds2)
Scenario 3 (os3)	Scenario 11 (os11)	Scenario 3 (ds3)
Scenario 4 (os4)	Scenario 12 (os12)	Scenario 4 (ds4)
Scenario 5 (os5)	Scenario 13 (os13)	Scenario 5 (ds5)
Scenario 6 (os6)	Scenario 14 (os14)	Scenario 6 (ds6)

Pure-Tone Recordings – Equipment Characterization Files

A series of short duration, pure tone signal recordings were made and included as part of the OAKBAT dataset. These signals were created in case a future user of the OAKBAT data were ever to need to independently characterize some aspect of the exact recording equipment chain used in the digitization of the VTS's RF signals. The SDR, LNAs, and even the cable used during the GNSS RF recordings were used for these pure-tone recordings. The intention is to allow OAKBAT users to independently confirm or eliminate any suspected intrinsic hardware behaviors using these files, even years from now.

The signal is a pure L1 (1.57542 GHz) sinusoid signal generated by an Agilent E4438C with the enhanced phase noise performance and high-stability time base options. Four recordings each, at two different power levels, were made, where the parameters varied were the internal gain of the SDR used and the use of external gain. The two power levels used were -125 dBm and -110 dBm. At each power level four 20-second recordings were made: 1) a recording with no internal gain and no external gain, 2) a recording with no internal gain, and 57 dB of external gain (see Table 1 for the two LNAs used for this), 3) a recording with 31.5 dB of internal gain (WBX-40's maximum analog gain) and no external gain, and 4) a recording with 31.5 dB of internal gain and 57 dB of external gain, which are the exact gains used for the OAKBAT recordings.

OPEN DATA ACCESS

To provide interested users of OAKBAT with reliable, open, and highly available access to OAKBAT, yet with effective but simple traceability, the metadata files, VTS scenario and configuration, and additional documentation are hosted in a source controlled versioning repository accessible at https://github.com/oakbat. The source control layer allows definitive verification of what the latest version of metadata and configurations available from the repository are as well as allow users to determine the version of their local copy. Additionally, the metadata site provides a centralized location to find the digital object identifier (DOI) links, along with the access instructions, pointing users to the long-term data archive system that host the large RF data files. This allows for future additions to OAKBAT be made available and released, such as a variant using a "rural" or "suburban" propagation environment, as additional "series" which will be announced and tracked on the metadata site. A list of the unique DOI for each "series" exist and its changes are tracked via the source control system. The authors welcome public feedback and questions via this website to allow the community of users to benefit from questions and answers, suggestions, and even possible issues encountered and shared.

FUTURE WORK

To advance OAKBAT in the future, as well as to make it an even closer "twin" of TEXBAT, it is intended that the same series of RF recordings will be reproduced using a propagation environment other than "open-sky." This capability is under development by Orolia for the VTS so the signal strength "fades" due to the environment will be synchronized between the dual GSG-6 simulators in the VTS. It will allow the spoofing power advantage to be applied to the dynamically changing power levels produced as the "sky" simulator simulates propagation loses so the spoofer's power advantage will be applied to the dynamically changing signal levels of the "sky" simulator. In simpler language, when the strength of a sky unit's satellite changes the spoofing unit's power advantage will mirror that change plus the addition of its power advantage.

The overhead GNSS coverage for the locations simulated in the scenarios was used in determining the date and time to be simulated. When this selection was done it was focused only on the GPS constellation. When Galileo scenarios were produced, after this decision, it was noticed that the overhead Galileo distribution was not as advantageous at the chosen date and time as it was for the GPS satellites. The use of a common starting date and time was considered the greater priority at the time, but the benefits of choosing a time and day in which both GPS and Galileo satellites are well positioned overhead needs to be evaluated and the possible benefits of re-creating the current 16 OAKBAT recordings should be assessed.

CONCLUSION

OAKBAT has been designed to supplement TEXBAT by providing a second, independent collection of data to enable the affordable advancement and verification of GNSS experiments. Most importantly, OAKBAT contains datasets that use the same key parameters of the TEXBAT ds1 through ds6 sets. By using these "sibling" datasets it is possible to cross-reference performance of algorithms and designs. This comparison will help in the objective evaluation and characterization of behavior not possible previously with only TEXBAT as a data source, such as assessing if performance is specific to some intrinsic behavior or properties unique to the hardware used to create a particular dataset.

Overall, OAKBAT results in a new resource for the community of researchers and developers working in the field of GNSS. This additional source of data provides a GNSS "playground" on which to experiment, explore, and evaluate new ideas. The combination of OAKBAT and TEXBAT provide a substantial tool which can help bridge the gap in resources between an initial idea and one meriting a production-level GNSS data source as well as providing a resource useful for educational purposes. It is the authors hope that by providing this dataset via a public open access platform it will encourage community involvement, improvement, and feedback to further advance the OAKBAT battery.

ACKNOWLEDGMENTS

Research sponsored by the Laboratory Directed Research and Development Program of Oak Ridge National Laboratory, managed by UT-Battelle, LLC, for the U. S. Department of Energy.

Austin Albright would like to thank David Lorah at Orolia USA for his patient support and assistance as we learned how to utilize all the features and capabilities of our Orolia VTS unit to achieve fully reproducible runs of the simulation system.

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